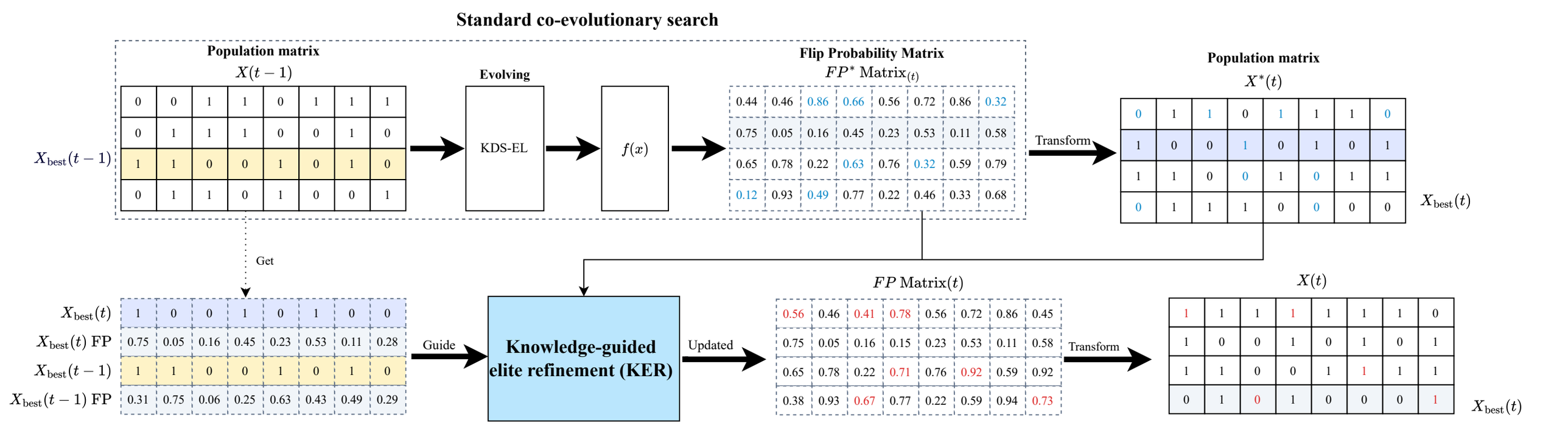

Standard co-evolutionary search
Population matrix
$X(t-1)$
Evolving
KDS-EL
$f(x)$
Flip Probability Matrix
FP^* Matrix$_{(t)}$
Transform
Population matrix
$X^*(t)$
$X_{\text{best}}(t-1)$
$X_{\text{best}}(t)$
Get
$X_{\text{best}}(t)$
$X_{\text{best}}(t)$ FP
$X_{\text{best}}(t-1)$
$X_{\text{best}}(t-1)$ FP
Guide
Knowledge-guided elite refinement (KER)
Updated
FP Matrix(t)
Transform
$X(t)$
$X_{\text{best}}(t)$